Bhavana Nare

AI Platform & Infrastructure Engineer

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PROFESSIONAL SUMMARY

AI Platform & Infrastructure Engineer with 11+ years designing and operating the cloud infrastructure, distributed systems, and developer platforms that scale ML and LLM/agent workloads. Proven track record architecting internal Agent SDKs, self-service developer platforms, and infrastructure-as-code (Terraform, Kubernetes/EKS) that reduced engineering ramp-up time by 40% and were adopted across 5+ engineering teams. Deep hands-on expertise across AWS (Bedrock, EKS, Lambda, SQS, EventBridge, CloudWatch, Redshift, S3), Azure, and GCP, with production experience in workflow orchestration (Temporal, Kubeflow, Airflow, MLflow), distributed tracing, and p95 SLO-driven observability. Combines platform engineering rigor with a user-first approach to building tooling that ML engineers, researchers, and data scientists rely on daily.

CORE COMPETENCIES

Cloud & Infrastructure: AWS (Bedrock, EKS, Lambda, SQS, EventBridge, CloudWatch, Redshift, S3) · Azure (Azure ML, Monitor) · GCP · Kubernetes · Docker · Terraform (IaC) · CI/CD (GitHub Actions)

Platform & SDK Engineering: Internal Developer Platforms · Agent SDK Design · Developer Frameworks & Tooling · GraphQL · RESTful APIs · Microservices · Event-Driven Architecture · Distributed Systems

Workflow Orchestration: MLflow · DAG orchestration · EventBridge-driven ML pipelines

AI / LLM / Agent Infrastructure: LangChain · LangGraph · AWS Bedrock (Claude) · RAG Pipelines · Agentic Systems · MCP / Tool-Calling · Plan-Act-Observe Loops · CrewAI

ML Platform & MLOps: ML Lifecycle Management · Experiment Tracking · Model Deployment · Training Infrastructure · Feature Pipelines · MLflow-compatible Eval Tracing

Observability & Reliability: Distributed Tracing · LangSmith · Structured Logging · p95 SLOs · Quality Signal Dashboards · Regression Harnesses · Grafana

Languages & Data: Python (FastAPI, Flask) · Java (Spring, SQS/EventListener) · Node.js · TypeScript · SQL · Apache Iceberg · PostgreSQL · MongoDB · Elasticsearch/OpenSearch

Frontend & Engineering Practices: ReactJS · TypeScript · TDD (pytest, Jest) · Design Patterns (MVC, Repository, Factory, Observer) · Agile/Scrum · Code Review

PROFESSIONAL EXPERIENCE

AI Senior Advance Tech Developer | ADP

Sep 2026 – Present, Georgia, USA

- Architected and built automation tools to streamline end-to-end system and framework upgrades with zero manual downtime.
- Engineered autonomous AI agents as part of ADP Assist, enabling multi-step workflow automation and generative AI reasoning across enterprise HCM systems.
- Designed deterministic guardrails, tool orchestration, and automated validation pipelines for reliable, production-grade agent execution.
- Integrated multi-agent architectures and LLM reasoning loops to modernize legacy upgrade procedures.

AI Platform & Agent Infrastructure Engineer | Fannie Mae

Apr 2026 – Sep 2026· Virginia, USA

- Architected production LLM agent infrastructure (Financial Insight Agent, Workflow Automation Agent, Engineering Productivity Agent) on AWS Bedrock, exposing enterprise data sources — Redshift, Apache Iceberg, internal APIs — as structured, tool-callable agent interfaces via plan-act-observe loops and MCP-style function calling.
- Built and scaled an internal Agent SDK adopted by 5+ engineering teams, standardizing agent deployment patterns and cutting time-to-market for new LLM-powered services.
- Engineered Java-based event-driven infrastructure (AWS SQS EventListener consumers) to trigger real-time ML pipeline execution, enabling automated retraining and downstream feature pipeline orchestration across distributed services.
- Designed end-to-end ML training infrastructure — feature ingestion (Apache Iceberg) through job orchestration, experiment tracking, and versioned model registration — supporting reproducible development across 3+ ML teams.
- Built a custom distributed tracing layer (AWS CloudWatch + structured JSON logging) capturing agent plan steps, tool invocations, and LLM latencies, establishing p95 latency SLOs and fast incident triage for non-deterministic agent paths.
- Authored eval suites and regression harnesses (LangSmith-compatible tracing) to measure task completion rates and detect quality regressions across prompt/model updates, driving quality signals consumed by 3 teams.
- Engineered scalable ETL/ELT infrastructure (Lambda, EventBridge, S3) and Elasticsearch indexing for full-text search, plus ReactJS natural-language query interfaces over Redshift and Iceberg datasets.
- Led TDD and design pattern adoption across backend services, reducing production defect rate and establishing team-wide code review and testing standards.

AI Engineer | Rivian Automotive
May 2025 – Mar 2026 · Michigan, USA

- Architected and shipped an internal Agent SDK and self-service developer platform (Python/FastAPI backend, ReactJS UI) enabling 5 engineering teams to build, onboard, and run LLM-powered agents in production — reducing ramp-up time by ~40% through opinionated scaffolding, API contracts, and documentation.
- Built a production-grade RAG agent infrastructure with tool-calling integrations (HuggingFace + OpenSearch Serverless), reducing expert escalations by ~50% on internal security policy queries.
- Implemented a Node.js/Express middleware layer for GraphQL schema federation, defining strict API contracts across agent tool interfaces, reducing over-fetching by 35% and enabling real-time developer dashboards.
- Established agent observability and evaluation practices: authored pytest/Jest suites with coverage thresholds, built tracing instrumentation for agent tool-call chains, and coached 4 junior engineers on AI-assisted development workflows.
- Integrated model versioning and deployment workflows into the developer platform, standardizing rollback procedures and reducing manual deployment steps across teams.
- Provisioned full infrastructure-as-code (Terraform) for distributed agent backend services on Kubernetes and presented reliability metrics and agent quality signals to senior leadership.

MLOps Engineer | Robert Bosch
Aug 2023 – Apr 2025 · Michigan, USA

- Implemented MLflow-based experiment tracking for Azure ML training runs, enabling data scientists to compare runs and promote best-performing models to staging with a single command.
- Built feature engineering pipelines feeding Azure ML training jobs, including automated data validation, schema checks, and retraining triggers on data drift detected via Azure Monitor.
- Built a scalable Azure ML pipeline predicting CI/CD risk at 85% precision, designing RESTful API endpoints (Python/FastAPI) serving predictions to Node.js services in a distributed systems context.
- Designed reusable GitHub Actions CI/CD pipeline templates following TDD standards, adopted by 3+ teams via written guides and self-service documentation — cutting release cycles from 3 days to 4 hours.
- Built Grafana + Azure Monitor observability dashboards with automated retraining triggers and communicated platform reliability improvements to global engineering teams.
- Developed internal developer productivity dashboards (React) with a focus on UX, accessibility, and performance optimization.

Senior Software Engineer – Computer Vision & Full-Stack | Continental Automotive
May 2019 – Jul 2021 · India

- At Continental, I worked in the Advanced Driver Assistance Systems (ADAS) domain, primarily supporting Camera Object Detection (COD). The camera-based object detection results were used along with radar information as part of the broader sensor-fusion and validation process.
- My responsibilities extended across software development, automation framework development, requirements analysis, testing, CI/CD, tool architecture, and Agile/Scrum leadership. I worked with development, testing, requirements, and customer-facing teams and took on multiple responsibilities depending on project needs.

AUTOSIM – ADAS Test Automation Framework

Role: Framework Developer, Tool Architect and Technical Specialist

- One of my most significant contributions was the design and development of AUTOSIM, a Python-based automation framework created to prototype and automate the testing process for Camera Object Detection components.
- I contributed to the framework from the architectural and prototype stages and designed it to be modular and configurable so that the approach could be extended to different components rather than creating separate automation for every use case.
- AUTOSIM automated multiple activities that previously required manual engineering effort. The framework executed component-level validation, generated reports using predefined templates, and automatically generated CSV files containing test-case information that could subsequently be uploaded into DOORS for each release.
- I integrated Jenkins to support automated execution as part of the CI/CD process and used Git for source-code management. JIRA was used for project tracking, while Confluence was used for technical documentation and retrospective information.
- The implementation reduced manual testing and validation effort by approximately 60%, while establishing a repeatable automated process for component validation and release reporting.
- My responsibilities went beyond framework implementation. As a Tool Architect and Technical Specialist, I participated in defining the technical approach, prototyping the solution, designing the framework structure, resolving technical issues, and supporting its adoption for ADAS component validation.

Camera Object Detection – Requirements and Validation

Role: Developer / Requirements & Validation Contributor / Scrum Master

- As part of the Camera Object Detection component team, I participated in monitoring and analyzing requirements and discussing requirements with different stakeholders and customers.

- For new or modified customer requirements, I supported requirement updates, developed and linked corresponding test cases, executed validation, and linked test results to the appropriate requirements. This provides hands-on experience with requirements of traceability and the automotive development lifecycle using DOORS, V-model practices, Agile methodologies, and ASPICE-related engineering processes.
- I also worked with the Camera Object Detection pipeline to understand model selection, training, testing, object-detection accuracy, and interpretation of model outputs within the ADAS application.
- In addition to my engineering responsibilities, I served as a Scrum Master when required, supporting sprint planning, backlog coordination, team discussions, retrospectives, and delivery activities.

ADAS-COD Visualization KPI Portal

Role: Developer

- I developed a KPI visualization portal using Python, Dash, Flask, Plotly, and SQLite to visualize Camera Object Detection performance and validation information.
- The portal transformed stored KPI data into graphs, tables, and other visualizations, allowing engineering teams to analyze results more efficiently. I worked with the development team to integrate, test, and validate the portal as part of the broader automation workflow.

IMS Dashboard

Role: Developer

- I also developed a web-based dashboard using Python and Django for viewing and analyzing tasks maintained within the IMS system.
- The application provided graphs, charts, tables, and automated email notifications for tasks approaching their due dates. Different views were created for stakeholders including managers, team leads, and customers. I collaborated with developers, testers, and requirements engineers during its development and used Git for version control and Confluence for documentation.

Significance and Impact

- My Continental role represented a significant progression in my professional responsibilities. I was not limited to implementing assigned software features; I worked across tool architecture, automation framework development, ADAS validation, requirements engineering, CI/CD, visualization, and Agile leadership.
- The most significant contribution was AUTOSIM, where I served as a Framework Developer, Tool Architect, and Technical Specialist and helped take the solution from prototype and architecture through implementation and automated deployment/testing. The framework reduced manual validation effort by approximately 60% and automated report and test-case artifact generation for the release process.
- My responsibilities in the Camera Object Detection team also required interaction across requirements, development, and testing activities, while my Scrum Master responsibilities provided additional technical and team leadership experience.
- Together, these contributions demonstrate significant performance through the creation of reusable automation, measurable reduction in manual engineering effort, technical architecture responsibility, cross-functional ADAS engineering, requirements ownership, and team leadership.

Python Full-Stack Developer | Teradata India Pvt. Ltd.

Aug 2018 – May 2019 · India

- At Teradata, I worked on PyTeradata, a tool designed to provide a Python interface for SQL analytical functions available on the Teradata database platform. My work focused on simplifying and automating how these analytical functions could be accessed, configured, and tested through Python.
- A significant contribution was developing functionality that used JSON-based definitions to automatically generate API structures and outlines for Python test files. Java was used as part of the API-generation tooling, while Python was used for interacting with and validating the analytical functionality. This reduced the need to manually create repetitive API and test structures when supporting different analytical functions.
- I developed functional and unit test cases and worked extensively with SQL queries against the Teradata database to validate functionality and expected results. Existing database tables and reference datasets were used to exercise and validate the analytical functions.
- As part of this work, I gained hands-on experience with Teradata analytical functions including NTree, Decision Tree, K-Nearest Neighbors (KNN), and Confusion Matrix. My responsibility was not to develop these underlying algorithms themselves, but to build and validate the software interfaces and automated testing capabilities that enabled these existing database analytical functions to be accessed and tested through the PyTeradata framework.
- The development environment included Python, Java, JSON, SQL, Teradata Database, Git, and automated unit/functional testing, with the project following Agile and Scrum development practices.

Significance of the Contribution

- My contribution went beyond writing individual test cases or SQL queries. I worked on a reusable framework that used metadata-driven JSON definitions to automate portions of API and Python test-code generation for multiple analytical

functions. This approach allowed common functionality to be reused as additional analytical functions were incorporated rather than requiring each interface and its associated test structure to be created entirely manually.

- The role also required me to understand how different analytical functions consumed database tables, parameters, and input data and to translate those requirements into Python interfaces, JSON definitions, SQL validation, and automated tests.
- This experience represented a progression from the deployment and configuration automation work I performed at TCS to the development of reusable software tooling for enterprise database analytics at Teradata. It further demonstrated my ability to design automation frameworks, work across Python, Java, SQL, and database technologies, and develop reusable engineering solutions rather than limiting my responsibilities to routine application development.

Senior Software Engineer | Tata Consultancy Services (TCS)
Jun 2014 – Aug 2018 · India

AIT – Automated Installation Tool | 2014–2016

- I worked on the AIT tool, which was used to install software components on physical nodes. I contributed to the development using C, developed automated test cases, and configured Jenkins to improve continuous integration and automated testing.
- My responsibilities covered development, testing, maintenance, and technical support, providing experience across multiple stages of the software development lifecycle.

ESM – System Model Generation | 2016–2018

- Worked on the design and development of ESM, a system-model generation tool used to define the software and configuration required for deployment environments. The generated system model specified the required RPM packages, cluster size, and services that needed to be active for a given cluster. The resulting system model was subsequently provided as input to KIWI for the creation of deployable system images.
- My contribution extended beyond maintaining an existing application. I was involved in developing new tools and capabilities from the prototype stage, including analyzing new requirements, creating Proof-of-Concept (POC) implementations to evaluate technical feasibility, designing the solution, implementing functionality, and developing automated tests using Python and the unit test framework.
- For new and time-sensitive client requirements, I worked on rapidly converting requirements into functional solutions and supporting them through development, testing, and delivery. I received appreciation emails from project stakeholders for quick turnaround, responsiveness to requirements, and timely delivery of requested functionality.
- This role expanded my responsibilities from individual development and testing into POC development, system-level design, requirements analysis, solution development from scratch, and client-focused delivery. The ESM experience subsequently contributed to my work on the CSM automation toolchain, where I continued developing reusable tools for configuration validation and automated system/ISO generation.

CSM Automation Toolchain | 2016–2018

- I also contributed to the design and development of a set of Python-based tools that automated the configuration and ISO-generation process. These tools accepted YAML-based system definitions and transformed them through multiple validation and configuration stages into ISO files that could be used for software/package installation.
- The toolchain included:
  - csmcli – command-line functionality for creating and updating YAML configuration.
  - csmlint – validation of YAML configuration before downstream processing.
  - csmconfig – generation of system configuration from validated input.
  - csm2iso – generation of deployable ISO files containing the required packages and configuration.
- For CSM, I was involved in developing capabilities from scratch, including evaluating new requirements, creating POCs to establish technical feasibility, designing the implementation, developing and testing the solution, and supporting its integration into the overall deployment workflow.
- I also worked directly on the requirements needed for client deliveries. Some requirements had short delivery timelines, requiring rapid analysis, implementation, testing, and delivery. I successfully completed multiple such requirements and received appreciation emails for quick delivery and responsiveness in developing client-requested functionality.
- The projects used Python, YAML, Git, Jenkins, Linux, Agile, and DevOps practices. In addition to my technical responsibilities, I served as a backup Scrum Master, assisting with sprint planning meetings, coordination of sprint activities, and sprint demonstrations.

Significance and Impact:

- My work during 2016–2018 represented a progression beyond routine development responsibilities. I was entrusted with developing new tools and capabilities from the POC and prototype stages, translating client requirements into working engineering solutions, contributing to system design, and supporting time-sensitive client deliveries.
- The CSM projects **require** ownership across the engineering lifecycle—from understanding requirements and establishing feasibility through POCs to design, implementation, automated testing, integration, and delivery.

- My ability to respond quickly to new client requirements and deliver working solutions within demanding timelines was recognized through appreciation emails from project stakeholders. I also took on additional team responsibilities as a backup Scrum Master, supporting sprint planning and demonstrations.
- These contributions demonstrate sustained significant performance at TCS, particularly during 2016–2018, through technical ownership, development of new tools from scratch, proof-of-concept development, client-focused delivery, and responsibilities extending beyond individual software implementation.

EDUCATION

Master of Computer Science (Thesis) · GPA: 3.7/4.0 · University of Georgia, Athens, GA · 2021–2023

Bachelor of Technology in Computer Science · GPA: 7.9/10 · Sree Vidyanikethan Engineering College, India · 2010–2014

SELECTED PUBLICATIONS

- A Novel Computational Framework of Robot Trust for Human-Robot Teams — IEEE ICRA, 2025
- Interpretable Deep Learning for Biological Age Prediction: A Counterfactual Approach — Atlantis Press, 2026
- Wildfire Prediction and Visualization: A Machine Learning Approach Using U.S. Data — Atlantis Press, 2026
- Flexplot: Enhancing Data Visualization in Statistical Modeling — IEEE, 2025
- Memory-Aware Scheduling in Constrained Environments — IEEE, 2025

CERTIFICATIONS

- Build a Computer Vision App with Azure Cognitive Services — Microsoft, Feb 2026
- Dataiku ML Practitioner | Advanced Designer | Core Designer — Dataiku, Mar 2026
- ISTQB Certified Tester Foundation Level (CTFL) — ITB, Dec 2020